

XSLT - XML Assignment:

1> sort based on rank that is mark high to low

```
<xsl:variable name="max">
  <xsl:for-each select="key('classKey', $cls)">
    <xsl:sort select="marks/math + marks/science + marks/english"
      data-type="number" order="descending"/>
    <xsl:if test="position()=1">
      <xsl:value-of select="marks/math + marks/science + marks/english"/>
    </xsl:if>
  </xsl:for-each>
</xsl:variable>
```

classKey — key for student/class

Cls — variable to use class (ie 9th or 10th)

2> Use <xsl:key> to look up grades from <grades> based on total.:

Instead of gradekey , class key is more necessary in this xml case so I used classkey for class lookup for 9th and 10th

```
<xsl:key name="classKey" match="student" use="class"/>
```

```
<xsl:variable name="max">
  <xsl:for-each select="key('classKey', $cls)">
    <xsl:sort select="marks/math + marks/science + marks/english"
      data-type="number" order="descending"/>
```

Class key is used to fast look up class 9 or 10 and use it in the necessary space

3> Flag top student (highest total) in bold/green.:

```
<xsl:for-each select="key('classKey', $cls)">
  <xsl:sort select="marks/math + marks/science + marks/english"
    data-type="number" order="descending"/>
  <xsl:if test="position()=1">
```

```
<style>
body { font-family: Arial; }
.top { background-color: lightgreen; font-weight: bold; }
```

Position 1 is mapped to the top of the table and top row is bold and green bg is added.

4> Compute/show class average total.:

```
<xsl:variable name="sumTotal"
  select="
    sum(key('classKey', $cls)/marks/math) +
    sum(key('classKey', $cls)/marks/science) +
    sum(key('classKey', $cls)/marks/english)
  "/>

<p>
  Average:
  <xsl:value-of select="format-number($sumTotal div count(key('classKey', $cls)), '0.00')"/>
</p>
```

Total sum is computed and then average is computed

5> template for grade:

```
<xsl:template name="getGrade">
  <xsl:param name="total"/>

  <xsl:value-of select="
    /school/grades/grade[@min &lt;= $total]
    [not(@min &lt; ../grade[@min &lt;= $total]/@min)]
  "/>
</xsl:template>
```

Get all grades select all valid grades (i.e. total grade 225 valid min grade range is 200,150,100,0) and biggest of the set is the valid grades. that is B

Output:

localhost:8000/acxhange-assignments-/xml

School Report

Class 9

Rank	Name	Total	Grade
1	Meena	274	A
2	Kavi	225	B

Average: 249.50

Class 10

Rank	Name	Total	Grade
1	Asha	255	A
2	Rahul	157	C

Average: 206.00

